

ARAUULLO UNIVERSITY

PHINMA EDUCATION NETWORK
COLLEGE OF INFORMATION TECHNOLOGY
CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

This chapter presents the background of the study and defines the problem addressed by the research. It outlines the study's objectives, scope, and limitations, as well as the significance of the research and the definition of key terms, providing a clear foundation for the succeeding chapters.

1.1 Background of the Study

LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City, Located at Bayanihan, Gapan City, Nueva Ecija. The Government Services Office of Gapan City, in coordination with a government organization, is responsible for managing and allocating government-owned heavy equipment and service vehicles for public use. These resources include ambulances and other heavy equipment that are borrowed by residents, organizations, events, and government offices for emergency response and community-related activities.

Since Mayor Emeng Pascual was elected Mayor in 2016, the responsibility for managing all heavy equipment has been formally assigned to the Government Services Office. As the demand for government services increased, the number of borrowing transactions also grew, making efficient resource management more important.

At present, the office relies on a manual, paper-based method for recording borrower information, schedules, and equipment usage. Staff members list all transactions on paper, and checking equipment availability requires reviewing records one by one. This process is time-consuming, prone to errors, and causes difficulty in monitoring which equipment is currently in use or available.

To address these challenges, the proponents proposed LingkodGapan: a Resource Allocation Management System that aims to automate the process of tracking, allocating, and managing government resources. The system is intended to reduce staff workload, minimize manual errors, organize records efficiently, and improve the delivery of government services to the people of Batang Gapan City, Nueva Ecija.

1.2 Statement of the Problem

The Government Services Office of Gapan City currently uses a manual logging process. In which records are listed on paper to manage the distribution of supplies, equipment and service vehicles. Because all records are written on paper, staff find it difficult to quickly know which equipment is available, who is currently borrowing, and when the equipment should be returned. This situation causes delays, added workload, and confusion in daily operations.

1. How may the proposed LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City be designed and developed using the Rapid Application Development (RAD) in terms of the following:

1.1 Requirements Planning

- Meeting with the staff of GSO to gather needs and identify problems in the manual logging process.

Determine required features such as tracking supplies, equipment, and vehicles.

1.2 Use Design

- Create simple screen layouts and system prototypes. Ask users to review and provide feedback to ensure the system is easy to use.

1.3 Construction

-Develop the system by coding, building the database, and testing functions like allocation, borrowing, and return tracking.

1.4 Cutover

-Convert paper records into the system and fully implement the LingkodGapan system to replace manual logging in the GSO office.

2. How may the IT experts evaluate the LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the following ISO/IEC 9126 Software Product Quality Standards criteria:

2.1 Functionality

-IT experts evaluate whether the system meets all required functions such as tracking supplies, managing equipment , and monitoring vehicle usage. They check if the system performs correctly and supports all necessary tasks of the Government Services Office.

2.2 Reliability

-IT Experts test how consistent and stable the system is. They check if the system works without crashing, handling errors properly, and ensures accurate data recording even during high usage.

2.3 Usability

-IT experts assess whether the system is easy to use. They evaluate the user interface, navigation, and clarity of instructions to ensure staff can operate the system with minimal training.

2.4 Efficiency

-Experts evaluate how fast the system responds and how well it uses resources like memory and processing power. They check if the system can quickly process requests and handle multiple users without delays.

2.5 Maintainability

-IT experts review how easy it is to update or fix the system. They check the system's code structure, documentation, and ability to add new features or correct errors without major changes.

2.6 Portability

-Experts check if the system can be used on different devices or platforms. They evaluate whether the system can run smoothly on various computers or networks used by the government office.

3. How may the End-user evaluate the LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the following ISO/IEC 9126 Software Product Quality Standards criteria:

3.1 Functionality

-The end-users may evaluate the system based on its ability to perform required functions such as recording, tracking, and managing the borrowing and return of heavy equipment and service vehicles. The system should be able to provide accurate information on equipment availability, borrower details, and return dates.

3.2 Efficiency

-The end-users can assess the system based on the time it takes to process transactions and retrieve information. The system should be able to reduce processing time, avoid delays, and enhance work performance compared to the manual system.

3.3 Usability

-The end-users may evaluate the system based on its ease of use, simplicity of interface, and simplicity of navigation. The system should be easy to learn and use by the government employees, even if they lack technical knowledge.

4. How may the End-user evaluate the LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the System Usability Scale?

The end-users evaluate the LingkodGapan: Resource Allocation Management System using the System Usability Scale after using the system. The respondents include the super admin, admin, barangay

representatives, and operators. The result shows that the system is easy to use and understand for all four users. The admin finds the system helpful in tracking equipment, vehicles, and assigned operators. The barangay representatives can easily send requests for equipment and vehicles. The operators can clearly see their assigned vehicles and schedules, while the residents find the borrowing process simple and organized. Overall, the System Usability Scale result indicates that the system is user-friendly, reduces manual logbook work, and helps improve the resource allocation process in the Government Services Office.

5. What are some comments and suggestions by the respondents for better improvement of the proposed subjects?

The respondents suggested that the proposed system can be improved by adding notification features for requests, approvals, and vehicle assignments. They also recommended improving the system interface to

make it easier to use for all users, especially residents. In addition, they suggested adding report and history features to better track borrowed equipment, vehicle usage, and maintenance records. Overall, these improvements can help make the system more efficient and reliable for the Government Services Office.

1.3 Objectives of the Study

The primary objective of this study is to design, develop, and evaluate a GSO Inventory and Procurement Management System (IPMS) for the Government Services Office of Gapan City using the Rapid Application Development (RAD) methodology.

Specifically, the system intends to accomplish the following:

1. To develop the proposed LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City be designed and developed using the Rapid Application Development (RAD) in terms of the following:

1.1 Requirements Planning

- Rapidly list key needs through group talks, like tracking stock and buying items.

1.2 Use Design

- Build interactive prototypes and UI mockups with user feedback to refine key workflows.

1.3 Construction

- Develop the full system incrementally through coding sprints and ongoing user testing.

1.4 Cutover

- Deploy the system with training, data migration, and live support for seamless transition.

2. To evaluate by the IT experts the proposed LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the following ISO/IEC 9126 Software Product Quality Standards criteria:

2.1 Functionality

- Complete, accurate features for inventory and procurement.

2.2 Reliability

- Fault tolerance and consistent performance.

2.3 Usability

- Ease of learning and operation.

2.4 Efficiency

- Optimal time response and resource use.

2.5 Maintainability

- Ease of updates and testing.

2.6 Portability

- Adaptability across environments.

3. To evaluate by the End-users the proposed LinkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the following ISO/IEC 9126 Software Product Quality Standards criteria:

3.1 Functionality

- Meets daily inventory/procurement needs.

3.2 Efficiency

- Speed in real workflows.

3.3 Usability

- Self-explanatory or easy to use.

4. To evaluate the End-users the proposed LingkodGapan: Resource Allocation Management System for the Government Services Office of Gapan City based on the System Usability Scale.

5. To evaluate the comments and suggestions by the respondents for better improvement of the proposed subjects.

1.4 Scope and Limitations

1.4.1 Scope of the Study

This research is specifically designed to address the operational needs of the Government Services Office of Gapan City. Its primary goal is to develop and deploy "LingkodGapan: A Resource Allocation Management System" to modernize how the office tracks and manages its fleet of government-owned assets, such as heavy equipment and service vehicles (e.g., ambulances, dump trucks).

The scope of this work is defined across three key areas:

1. **Core Functionality:** The system will digitize the entire equipment management process. This includes:
 - Maintaining a centralized digital registry of all assets and authorized borrowers (e.g., barangay offices).
 - Enabling staff to monitor real-time equipment availability, manage borrowing schedules, and log returns.
 - Storing all transaction records in a secure, searchable database and generating basic reports for operational oversight.
2. **Defined User Roles:** Access to the system is carefully structured around four user types to ensure security and clarity of responsibility:
 - **Super Admin:** Has complete oversight and system configuration authority.

- Representative (Admin): Manages daily operations, including equipment lists, borrower profiles, and transaction approvals.
 - Barangay (User): External officials who can view availability and submit borrowing requests online.
 - Operator (Staff): On-ground personnel who process physical check-outs/check-ins and update equipment status.
3. Project Boundaries: It is important to note that this phase of the project is focused solely on creating the core management software for the Gapan City office. The study does not cover hardware integration (like GPS trackers), financial modules for billing, or long-term impact assessments post-deployment.

1.4.2 Limitations of the Study

While the system aims to significantly improve operational efficiency, this study acknowledges several inherent constraints that define its current capabilities and potential challenges.

- Technical and Operational Constraints: The system is a web-based platform designed exclusively for the Gapan City Government Services Office. Its functionality is therefore contingent on a stable internet connection and cannot be directly transferred to other municipalities without substantial redevelopment.
- Access and Adoption Barriers: The shift to a digital platform naturally introduces a digital divide. The system assumes users have a basic level of digital literacy and device access, which may unintentionally disadvantage elderly or low-income residents, as well as those in areas with unreliable internet.
- Structural and Human Factors: The system operates on a fixed four-role architecture (Residents/Users, Drivers/Operators, GSO Admin, Mayor's Office). While this ensures clear boundaries, it may lack the flexibility for granular permission controls or to easily adapt to future changes in staff duties or

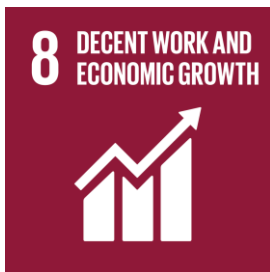
collaborative workflows. Furthermore, maintaining the system creates a long-term dependency on specific technical expertise, as it relies on developers familiar with its custom architecture.

- **External Dependencies:** Certain features that enhance user experience, such as SMS notifications or online payment processing, depend on third-party services. The system's reliability in these areas is therefore partially subject to the uptime and policies of these external providers, which are outside the project's direct control.

1.5 Significance of the Study

The proposed project solutions aim to target the following Sustainable Development Goals (SDGs):

ADD 3 SDGs based on your study



Decent Work and Economic Growth : Enhancing the working conditions of the employees in the Government Services Office of Gapan City. The system is an improvement over the manual process because it reduces the workload of the employees and enables them to work more efficiently. The system is beneficial to government employees because it makes their work easier, faster, and more organized. The efficient management of heavy equipment and service vehicles ensures that government resources are utilized properly for the delivery of public services. The system is beneficial to local employees, contractors, and the community because it supports the operations of the city of Gapan.

Sustainable Cities and Communities : Improving the management of equipment used for city services such as road maintenance, construction, and emergency response in Gapan City. This is important because it enables the city government to respond more quickly to community needs. This directly benefits city residents who rely on timely and effective public services. Efficient resource allocation reduces delays and service conflicts, leading

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to better coordination among government offices. This benefits the community by promoting safer, more organized, and more sustainable services within the city.



Responsible Consumption and Production : It aims to ensure responsible use of government-owned equipment and service vehicles. Proper monitoring of borrowing, usage, and return schedules will help prevent misuse and unnecessary damage to public resources. This will benefit the local government by ensuring that public assets are

properly maintained and used efficiently. The conversion from a paper based system to a digital system will help reduce the use of paper and support environmentally responsible practices. This benefits both the government and the environment by minimizing waste and promoting sustainable operations in the delivery of public service.

1.5 Definition of Terms

Resource Allocation

Operational: Refers to the process within the LingkodGapan system where admin assign available equipment to approved borrowers and record the schedule of use and admin will assign who will be the driver.

Conceptual: The process of distributing government-owned equipment and vehicles to individuals or organizations that need them for public service and community activities.

Resource Allocation Management System

Operational: Refers to the list of government-owned equipment and service vehicles encoded and monitored in the system, including ambulances and other service related equipment.

Conceptual: An integrated system for managing the encoding, tracking, and monitoring of government-owned resources like equipment and vehicles.

Borrower

Operational: Refers to the person or group whose information is encoded in the system when borrowing equipment, including their name, purpose, and borrowing schedule.

Conceptual: Any individual, organization, or government office that requests and uses government owned equipment for a specific time and purpose.

User-Friendly System

Operational: Refers to the design of the LingkodGapan system that allows admin to easily input data, check availability, and manage records without technical difficulty.

Conceptual: A system that is easy to understand, allocate, and use, allowing staff to perform tasks efficiently with minimal difficulty.

Heavy Equipment

Operational: Government-owned vehicles including ambulances and other service equipment encoded and monitored in the LingkodGapan system for allocation.

Conceptual: Government owned vehicles, including ambulances and other service equipment, used for emergency response and community services.

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